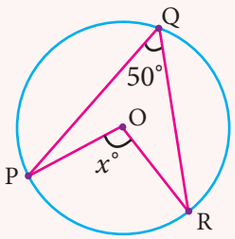
Solution

Using the theorem the angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the remaining part of a circle.

(i) $\angle POR = 2\angle PQR$

$x^\circ = 2 \times 50^\circ$

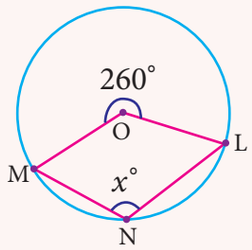
$x^\circ = 100^\circ$



(ii) $\angle MNL = \frac{1}{2}$ Reflex $\angle MOL$

$= \frac{1}{2} \times 260^\circ$

$x^\circ = 130^\circ$



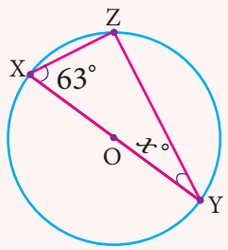
(iii) XY is the diameter of the circle.

Therefore $\angle XZY = 90^\circ$
(Angle on a semi – circle)

In ΔXYZ

$x^\circ + 63^\circ + 90^\circ = 180^\circ$

$x^\circ = 27^\circ$



(iv) $OA = OB = OC$ (Radii)

In ΔOAC,

$\angle OAC = \angle OCA = 20^\circ$

In ΔOBC,

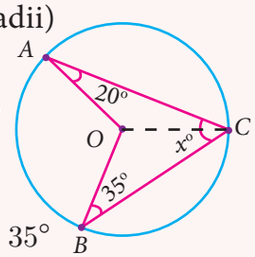
$\angle OBC = \angle OCB = 35^\circ$

(angles opposite to equal sides are equal)

$\angle ACB = \angle OCA + \angle OCB$

$x^\circ = 20^\circ + 35^\circ$

$x^\circ = 55^\circ$



Example 4.8 If O is the centre of the circle and $\angle ABC = 30^\circ$ then find $\angle AOC$. (see Fig. 4.73)

Solution

Given $\angle ABC = 30^\circ$

$\angle AOC = 2\angle ABC$ (The angle subtended by an arc at the centre is double the angle at any point on the circle)

$= 2 \times 30^\circ$

$= 60^\circ$

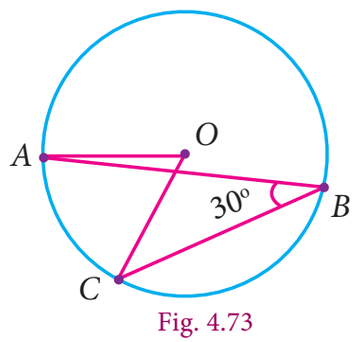


Fig. 4.73

Now we shall see, another interesting theorem. We have learnt that minor arc subtends obtuse angle, major arc subtends acute angle and semi circle subtends right angle on the circumference. If a chord AB is given and C and D are two different points on the circumference of the circle, then find $\angle ACB$ and $\angle ADB$. Is there any difference in these angles?

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